📘 MODULE 28: ALERTS

28.1 Search vs Saved search vs Report vs Alert (memorize)

Term	Meaning
Search	Something you type and run once
Saved search	A search stored with a name (the underlying object for the next two)
Report	A saved search whose **results are displayed or scheduled** for viewing/emailing
Alert	A saved search that is **scheduled (or real-time)** and **triggers an action** when a condition is met
Dashboard	A page of panels. Panels can be backed by reports/searches.

In `savedsearches.conf`, all of them are stanzas of the same file.

28.2 Creating an alert

1. Run your search and set the time range.
2. **Save As → Alert.**
3. Fill in:
   - **Title**
   - **Alert type:** *Scheduled* or *Real-time*
   - **Schedule:** run every hour / day / **cron expression**
   - **Time range** for the search (should usually match the schedule interval)
   - **Trigger condition:** *Number of Results* (is greater than 0), *Number of Hosts*, *Number of Sources*, or *Custom*
   - **Trigger:** *Once* or *For each result*
   - **Throttle:** suppress repeats (for example, 60 minutes by a field such as `src_ip`)
   - **Actions:** Add to Triggered Alerts, Send email, Webhook, Run script (where available), Log event
4. **Save.**

28.3 Scheduled vs real-time

	Scheduled	Real-time
Runs	At cron intervals	Continuously
Cost	Low	High (holds resources constantly)
Best for	Most alerts	Rare cases; production teams usually avoid it

Best practice: scheduled alerts with a short interval (e.g. every 5-15 minutes) cover most needs.

28.4 Cron cheat sheet

Cron	Meaning
`*/5 * * * *`	Every 5 minutes
`0 * * * *`	Every hour
`0 8 * * 1-5`	08:00 on weekdays
`*/15 * * * *`	Every 15 minutes

Format: `minute hour day-of-month month day-of-week`.

28.5 Worked alert: brute force

Query

```spl
index=linux_auth sourcetype=linux_secure "Failed password"
| rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count, dc(user) AS users_tried by src_ip
| where count>=10
```

What it does: finds source IPs with 10 or more failed logins in the search window.
Alert settings:

- Schedule: every 15 minutes (`*/15 * * * *`), time range **last 15 minutes** (`earliest=-15m`).
- Trigger: number of results > 0, **for each result**.
- Throttle: 60 minutes, by `src_ip`, so the same IP doesn't alert every 15 minutes.
- Action: email, or add to Triggered Alerts.

Common mistakes:

- Time range doesn't match the schedule (gaps or overlaps).
- No throttle, which causes **alert fatigue**.
- Threshold never tested against real data.
- Email isn't configured (Settings → Server settings → Email settings, which needs an SMTP server).

Interview explanation: "I schedule the search every 15 minutes over the previous 15 minutes, trigger per result, and throttle by source IP so analysts get one alert per attacker, not 4 per hour."

28.6 As configuration (`savedsearches.conf`)

```ini
```

```
[Brute Force Alert]
search = index=linux_auth sourcetype=linux_secure "Failed password" | rex "from (?<src_ip
enableSched = 1
cron_schedule = */15 * * * *
dispatch.earliest_time = -15m
dispatch.latest_time = now
counttype = number of events
relation = greater than
quantity = 0
alert.suppress = 1
alert.suppress.period = 60m
alert.suppress.fields = src_ip
alert.track = 1
```

Setting names are standard, but keep this as a reference. Create it in the UI first, then read the generated file.

28.7 Severity

Alert severity (Info, Low, Medium, High, Critical) is a **label** for triage, and it doesn't change how Splunk runs the search. In ES, severity feeds urgency.

28.8 Webhooks (concept)

A webhook alert action sends an HTTP POST with the alert details to a URL (a chat tool, ticket system, SOAR). Splunk doesn't need to know the target's internals.

28.9 Hands-on

1. Create an alert on `index=_internal sourcetype=splunkd log_level=ERROR` that triggers if more than 0 results in the last 15 minutes, every 15 minutes. Action: **Add to Triggered Alerts**.
2. Cause a trigger (for example, restart Splunk, or add a failing input) and check **Activity → Triggered Alerts**.
3. Add a throttle on `component`.
4. Check how it ran:

spl

```
index=_internal sourcetype=scheduler savedsearch_name="<your alert>" | table _time sta
```

28.10 Troubleshooting: “alert does not fire”

Check	Fix
Does the search itself return results for the window?	Run it manually with the same time range
Is it scheduled and enabled?	Settings → Searches, reports, and alerts. Enable it.
Scheduler log: `status=skipped`?	Concurrency limit hit, so stagger or reduce searches
Trigger condition too strict	Check threshold and trigger type
Throttle suppressing	Check the suppression window and field
Email action failing	Check SMTP settings and `index=_internal sourcetype=splunk_python OR source=*python.log* sendemail`
Time range/latency	Events arrive late (`_indextime` ≫ `_time`), so extend the window or use `_index_earliest`
Permissions	The alert’s owner must be able to search the index

28.11 Interview Q&A

- **Report vs alert?** A report displays results, and an alert triggers an action on a condition. Both are saved searches.
- **What is throttling?** Suppressing repeat alerts for the same value for a set period.
- **Why avoid real-time alerts?** They continuously consume resources, so scheduled searches are usually enough.
- **How do you reduce alert fatigue?** Throttle, tune thresholds, allowlists via lookups, and aggregate by entity.
- **Per-result vs once?** Per-result fires one action per row, and “once” fires once for the whole result set.

28.12 Revision

Search → saved search → report/alert · schedule via cron · time range matches interval · throttle by field · scheduled > real-time · debug via `sourcetype=scheduler`.